

Direct Detection of *Treponema pallidum*

Direct detection of the causative agent — *Treponema pallidum* — is an absolute criterion for the definitive diagnosis of syphilis.

T. pallidum may be detected in clinical specimens obtained from lesions or infected lymph nodes in cases of early syphilis using the following methods: - **Dark-field microscopy** - **Direct fluorescent antibody test (DFA-TP)** - **Nucleic acid amplification tests (NAATs)**, such as polymerase chain reaction (PCR)

Table 1: Laboratory Criteria for the Diagnosis of Syphilis

Diagnosis	Methods used
Definitive	Demonstration of <i>T. pallidum</i> in clinical specimens by dark-field microscopy, DFA-TP, validated NAAT, or equivalent method.
Presumptive	Positive result using two types of serologic tests: • A positive non-treponemal test (e.g., RPR, VDRL, MPR, TRUST) • A positive confirmatory treponemal test (e.g., TPHA, TPPA, ELISA, FTA-ABS)

Table 2: Sensitivity and Specificity of Non-Treponemal Tests for the Serological Diagnosis of Syphilis

Non-treponemal test	Primary syphilis (%)	Secondary syphilis (%)	Early latent syphilis (%)	Late syphilis (%)	Specificity (%)
MPR*	81 (70–90)	91	94 (88–100)	70 (57–80)	98 (93–99)
RPR	86 (77–100)	100	98 (95–100)	73 (57–85)	98 (93–99)
VDRL	78 (59–87)	100	95 (88–100)	71 (37–94)	98 (96–99)
TRUST	85 (77–86)	100	98 (95–100)	–	99 (98–99)

MPR: Microscopic particle agglutination test; used predominantly in countries of the Newly Independent States (NIS).

Dark-Field Microscopy

Dark-field microscopy can be used to diagnose primary and secondary skin lesions of sexually acquired syphilis, as well as early congenital syphilis. It is rarely used in tertiary syphilis, though it may be applicable if material is collected from the base of an active infiltrate.

Specimens for dark-field examination can also be obtained from: - Aspirates of regional lymph nodes - Cerebrospinal fluid - Amniotic fluid

However, the presence of non-pathogenic commensal treponemes — such as *Treponema refringens*, *Treponema phagedenis* (reiteri) in the urogenital tract, and *Treponema denticola* in the oral cavity — complicates interpretation when lesions are located in the mouth or rectum, due to morphological similarity to *T. pallidum*.

For specimens from oral or rectal sites, **DFA** or a **NAAT** is preferred over dark-field microscopy.

Direct Fluorescent Antibody Test (DFA)

DFA is not widely used in Eastern Europe due to limited availability of commercial reagents, particularly FITC-labeled monoclonal antibodies specific for *T. pallidum*.

Nucleic Acid Amplification Tests (NAATs)

Currently, there are no commercially available NAATs that detect *T. pallidum*-specific DNA sequences. However, numerous in-house NAATs — including real-time PCR assays — have been developed and utilized in research and reference laboratories.

All in-house NAATs must be rigorously validated.

Ideal validation should include testing of at least **30 specimens** from patients with:

- Dark-field microscopy-positive lesions
- Confirmed seropositivity for syphilis